# The Open Doors of a Dark-Energy-Tied Galaxy Law: Scoped No-Gos, Record Corrections, and a Cassini Split between the Unscreened Law and the Screened Theory

Working note, 2026-10-22, v1.1 (corrects the native-$H_0$ statements of v1.0; see the end). AI-assisted research programme; not peer reviewed. $\kappa = \frac{1}{2}$ is fitted; the theory is not closed.


**Abstract**

The owner's programme studies a gravity framework whose galactic law is $g_N = g^2/(g+a_0)$ with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ fitted. Its extra gravitating mass is a collisionless "cold energy" medium, not a particle. On 2026-10-22 we made one rigorous pass over every door the record had left open (lanes D1–D9b), followed by an audit of hidden inputs (wave E1–E5). Each lane froze its criteria before computing and ran a mutation control. Both $a_0$ footings ($9.36\times10^{-11}$ and $1.13\times10^{-10}\,\mathrm{m\,s^{-2}}$) are reported and never pooled. Most results are scoped no-gos and corrections to the record. A finding that changes several others: the framework's own action gives AQUAL with $\mu = x/(1+x)$, not the QUMOND that three lanes used (E2).

1. **Cassini splits the theory in two.** The unscreened canonical law fails Cassini $Q_2$ by 2.44–4.59× the Hees et al. (2014) $2\sigma$ line (canonical) and 2.85–5.19× (alternative) with the action's own field equation (QUMOND: 1.70–3.70× and 1.91–4.03×). The strong-field turn-off cannot help: across its whole band it moves $Q_2$ by $0.052\sigma$. This is the RAR–$Q_2$ class conflict of Desmond, Hees & Famaey (2024). The screened theory (stiffness length $L_\star = 24$–$90\,\mathrm{pc}$) passes: its $Q_2$ is at most $2.5\times10^{-8}$ of the Hees line, because inside $L_\star$ the Sun's field is a hedgehog with no quadrupole; it passes for every $L_\star$ above a threshold that is at most 0.069 pc (canonical) or 0.111 pc (alternative), an upper bound on the threshold.
2. **The screened theory is the one version that passes both Cassini and the galactic law.** Its open items, stated plainly: the satellite dwarfs sit $+0.89\pm0.13$ dex above its prediction, and the cold mass needed is supplied only through a ΛCDM-processed census with no binding mechanism in the record (E3); $L_\star$ is soft (native floor 28.85 pc, 20.0–42.3 pc over the stellar $M/L$ uncertainty) and $u$ is an $a_0$- and $h$-dependent number (0.433 at the galaxy-measured $a_0$; E1, E4); no covariant parent exists for the stated classes, with or without stiffness (D9, D9b, E5). Gaia DR4 is the decisive test: the screened theory predicts $\gamma_v - 1 \approx 8.1\times10^{-6}$ for a 20 kAU binary, the unscreened law-first reading $\gamma_v \approx 1.17$.
3. **$\kappa$ is not derived.** No horizon-thermodynamic monomial gives $\frac{1}{2}$ without square roots; with them, $\frac{1}{2}$ appears only as $\sqrt{G\rho_\Lambda}$ times a half picked from a menu. The class's natural output is Milgrom's $a_0 = cH_\Lambda/2\pi$, a postulate here. The data do not select $R = 4$ ($\kappa = \frac{1}{2}$) among natural caps: 5 of 7 grid values lie within $2\sigma$. The measured $\kappa$ is robust to the background (E4). On the background that BBN allows, the framework's native $H_0$ is 67.2–68.6 km/s/Mpc, 4.4–5.8$\sigma$ below the local ladder: the ordinary Hubble tension (C2; corrected in v1.1).
4. **The unscreened law family.** No member passes both $Q_2$ and SPARC on both footings under AQUAL (E2). In an $a_0$ bracket around the canonical value, with QUMOND, some members passed (E1); AQUAL raises their $Q_2$, so their status is knife-edge, not settled. The turn-off has no mechanism in the action, and its scale $y_t$ is $\kappa$ relabelled.
5. **No covariant parent.** A U(1)-dielectric parent that enforces the law's Gauss identity is superluminal or needs a dark-sector Lorentz violation of $6.3\times10^7$ at 1 AU; enforcing the identity in the record's healthy action brings back the no-self-gravity theorem; the stiffness changes none of this (E5).
6. **Tests, inputs and record.** The skew-sign test of the law's form is retired; the wide-binary fork is decided by systematics, not sample size; the sealed ALMA tests are expected to be unscorable; the native $1024^3$ shear run died of memory with no verdict. No tested mechanism gives the retained fraction $u \approx 0.47$ ($13.0\sigma$ from the record's cluster model; inflow stream counting is rejected by the profile, $\chi^2 = 39.0$ for 8 degrees of freedom). Four earlier verdicts flip under clean inputs (E1; one of them, D7c, only in part, see v1.1 note). Six errata were written into the record.


# 1 Setting and rules

- **The law.** $g_N = g^2/(g+a_0)$, the record's canonical ($n = 1$) form. A tethered cold medium, whose excess density is the law's phantom, supplies the extra mass (PAPER57). Footings: canonical $a_0 = 9.36\times10^{-11}$, alternative $1.13\times10^{-10}\,\mathrm{m\,s^{-2}}$.
- **$\kappa$.** $\kappa = \frac{1}{2}$ is fitted. The thermal value $\kappa = Z/4\pi = 0.460659$ ($a_0 = cH_\Lambda/2\pi$, $Z = 2\sqrt{8\pi/3} = 5.78881$) is a postulate. It is Milgrom's relation $2\pi a_0 \approx c^2(\Lambda/3)^{1/2}$ (arXiv:1605.07458, eq. 5), with the de Sitter Unruh mechanism of Milgrom (1999). The record's DSF-21 documents were corrected to say so (commit d3e4110528).
- **Field equation (E2).** Varying the record's action (PAPER49; DSF-21 A33 at $L_\star \to 0$) gives $\nabla\cdot[\mu(|\nabla\Phi|/a_0)\nabla\Phi] = 4\pi G\rho$ with $\mu = x/(1+x)$: AQUAL, not QUMOND. The action makes $\nabla\Phi$ curl-free but not

Table 1: The lanes of the doors push and of wave E. "Checks" is the main run; the mutation run must fail at least one. Commits: criteria / result. [a]Under QUMOND; see E1 and E2 for its status under clean inputs and the action's own field equation.

| lane | door | verdict | checks | commits |
|---|---|---|---|---|
| D1 | skew-sign test of $n = 1$ vs $n = 2$ | SCOPED NO-GO | 10/10 | 1f956de1e8 / ad9bea8355 |
| D2 | wide-binary fork | CRISP FINDING | 7/7 | c8960e6424 / c27b941b99 |
| D3 | CFG960 $1024^3$ run | CRISP FINDING (run) | – | 77d59a9a38 (read-only) |
| D4 | sealed ALMA tests | CRISP FINDING | 6/6 | 3537e6f817 / f81efdcf6f |
| D5 | why $\kappa$ | SCOPED NO-GO | 8/8 | 91ec2c9d07 / 962007715e |
| D6 | retained fraction $u$ | CRISP FINDING | 6/7 | 3082c43c79, A1 daf92d2d11 / b21dda1a2d |
| D7 | turn-off $y_t$ | SCOPED NO-GO | 11/11 | f2e0423a60 / a2a07b53d9 |
| D7b | Cassini $Q_2$, unscreened law | CRISP FINDING | 7/8 | c6a0b6f3c7 / 9542a09f54 |
| D7c | $Q_2$–RAR gate, unscreened family | SCOPED NO-GO[a] | 5/6 | e9e5349368 / 26b5db080f |
| D7d | Cassini $Q_2$, screened theory | CRISP FINDING | 10/10 | a13e418498, A1 e1ba64cf92 / b26f14bb82 |
| D8 | value of $L_\star$ | CRISP FINDING | 5/5, 2/2 | 3327019734 / 3d5eb3ead9, f1da138742 |
| D8b | universality of $\sigma_c$ | CRISP FINDING | 5/5 | 97ec7892fa / 8a9f0f1234 |
| D9 | covariant parent | SCOPED NO-GO | 9/12 + 3/3 | bfb93eb61c / 2edcee066c |
| D9b | A33 + multiplier | SCOPED NO-GO | 14/14 | ce12fb3210 / 61883a565f |
| E1 | input ledger and contamination | CRISP FINDING | 6/6 | 581d52210b / 77a8b2d4a6 |
| E2 | field equation of the action | CRISP FINDING | 10/10 | 021ee82abb / 474050bcd8 |
| E3 | dwarf satellite excess | CRISP FINDING | 6/6 | 5686635887 / c5e4196bb9 |
| E4 | the fitted three ($\kappa$, $L_\star$, $u$) | CRISP FINDING | 6/6 | 9cc8fcae21, A1 9b3dfb1000 / c5a9768e6a |
| E5 | parent logic with stiffness | SCOPED NO-GO | 14/15 | 039c76d2f9, A1 ab0f5d006c / 08e3c1bfbc |

$\mu\nabla\Phi$; the divergence-free part is 22% of the anomalous field at $r_M$. D2, D7b and D7c used QUMOND; where E2 recomputed them, the AQUAL values supersede the QUMOND ones and both are quoted.

- **The screened theory.** The same action with stiffness length $L_\star$ (the record's $L_\star \geq 24\,\mathrm{pc}$): galaxies follow the law, systems small next to $L_\star$ (the Sun, outer globulars, wide binaries) are screened toward Newton.
- **Method.** Each lane (directory `doors_push_2026_10_22/`) committed `FROZEN_CRITERIA.md` alone before any script, then a deterministic script whose output and `*_results.json` carry every number quoted here, and a mutation run that must break at least one check. Amendments are append-only and marked pre- or post-result. Table 1 lists the lanes.

## 2 Why $\kappa$? A no-go for the horizon class (D5)

The law gives $\rho_\Lambda c^2/(a_0^2/G) = 1/\kappa^2 \equiv R$, so $\kappa = \frac{1}{2}$ is $R = 4$, $\Lambda = 32\pi a_0^2/c^4$ and $a_0 = cH_\Lambda/Z$.

- **The class.** $a_0$ is any monomial in the de Sitter horizon quantities with their textbook coefficients (radius, area, volume, Gibbons–Hawking temperature, Bekenstein–Hawking entropy, Misner–Sharp, Komar and vacuum energies, surface gravity) times powers of $c, G, \hbar, k_B$, with the result free of $\hbar$ and $k_B$. Exponents are integers ($D = 1$) or half-integers ($D = 2$).
- **Theorem (checked on all 3,900 enumerated monomials).** Every member is $a_0 = q\pi^m cH_\Lambda$ with $q^2$ rational, so $\kappa^2 = (8/3)q^2\pi^{2m+1}$. For $D = 1$ the power of $\pi$ is odd, so $\frac{1}{2}$ is impossible, and $Z/4\pi$ is the class's simplest member ($a_0 = ck_BT_{\mathrm{GH}}/\hbar$). For $D = 2$ all 16 constructions that give $\frac{1}{2}$ have $m = -\frac{1}{2}$ and $q^2 = 3/32$: the law's own $\sqrt{G\rho_\Lambda}$ times one of three rational halves (Misner–Sharp, Komar, $\surd$ of Bekenstein's $\frac{1}{4}$).
- **No evidential weight.** The class covers the data band with probability 1.0000, for both classes and both bands, and for the 11-value minimal subclass. No member fixes a second, coefficient-dependent number: it reads either $H_\Lambda$ (constant $a_0$) or $H(z)$, giving $a_0(2.5)/a_0(0) = 3.7687$, the record's rival law. Nearest class values: 0.51824 (canonical) and 0.61421 (alternative) without square roots.
- **Record error.** The record says the data admit only $R = 4$ among natural caps. A19's own grid is $R \in$

$\{1, 2, 3, 4, 8, 12, 16\}$. On the CFG493 band ($\kappa = 0.417 \pm 0.095$), 5 of 7 lie within $2\sigma$, and $R = 8$ ($-0.67\sigma$) is closer than $R = 4$ ($+0.87\sigma$). Erratum e1fd7be4dc.

- **Scope.** Classes outside this one (bare-$G$ portals, media, quantised couplings) stay open. The data cannot rank $\frac{1}{2}$ against $Z/4\pi$.

# 3 The strong-field turn-off and Cassini

## 3.1 The turn-off has no mechanism, and $y_t$ is $\kappa$ (D7)

- **The canonical law needs a turn-off.** Its tail is $y(\nu - 1) \to 1$, a constant anomaly $a_0$. At 1.5 AU that is 2557.4 times the planetary bound $B_E$ on the canonical footing and 3087.4 times on the alternative. The Galaxy's external field cannot remove it: the monopole of the phantom source changes by less than $10^{-9}$.
- **$y_t$ is $\kappa$ relabelled.** The record's closure $I(y_t) = 8\pi/\kappa^2$ is one-to-one: $y_t = 66.2999$ (quoted $\Lambda$), 66.3803 ($32\pi$), 46.0148 (alternative), 77.876 (thermal). The falsifier band $y_t \in [60, 140]$ is exactly $\kappa \in [0.3415, 0.5267]$. Nothing measures $y_t$ now: at $y_t = 140$ Saturn's signal is $5.5 \times 10^{-4}$ of its bound, and wide binaries move by at most 0.57%.
- **No mechanism in the action.** The stiffness term is a mass threshold, not an acceleration threshold, and curing the planets that way is the medium-first reading. No gate potential $W(u)$ can produce the turn-off: the anomaly peaks at $y = 12.24$ with $u = 0.899\, a_0$, and the same $u$ recurs at $g = 0.816\, a_0$ and $72.9\, a_0$. None of 48 frozen scale monomials matches $y_t$ on all three points; a disclosed coincidence, 66.46, is off by 44% on the alternative footing.
- **Record errors.** The cure's necessity was not open (A4 had it); the "planets pass 9.14×" margin rests on a stand-in Neptune bound; an $O(10^2)$ external-field suppression cannot act on a $1/y$ tail; the thermal $y_t$ is 77.876, not 77.85 or 77.944. Erratum 3c11f5d2e0.

## 3.2 Cassini $Q_2$ of the unscreened canonical law (D7b, E2)

D7b computed the Solar-System quadrupole $Q_2$ of the canonical law in QUMOND, without stiffness screening (the DSF-21 A26 "law-first" reading), with and without the turn-off, for four external fields (A60's framework Milky Way, and kinematic totals 1.9, 2.2 and $2.6\, a_0$).

- **With the action's field equation (E2, AQUAL): the unscreened canonical law fails by 2.44–4.59×** the Hees line (canonical) and 2.85–5.19× (alternative), at every $y_t$ row; AQUAL/QUMOND is 1.23–1.49 depending on the external field. These values supersede D7b's QUMOND values below. E2's AQUAL solver reproduces Blanchet & Novak's AQUAL benchmark to 1.1%.
- **D7b (QUMOND, superseded): the unscreened canonical law fails Cassini by 1.7–4×.** Against the Hees et al. (2014) $2\sigma$ line ($9.0 \times 10^{-27}\,\mathrm{s}^{-2}$) it fails at every $y_t$, every external field and both footings: 1.70–3.70× (canonical) and 1.91–4.03× (alternative). It also fails the tighter Park et al. (2026) line. The most favourable field still fails (1.70× canonical).
- **The turn-off cannot help.** It changes $Q_2$ by +0.06% to +0.69% (upward). Across the band $y_t \in [60, 140]$, $Q_2$ moves by at most $1.6 \times 10^{-28}\,\mathrm{s}^{-2}$, $0.052\sigma$ of Hees et al. So no $\kappa$ range is selected. $Q_2$ is set near $y \approx 1$ ($r \approx 7000$ AU); the turn-off acts at $y > 60$.
- **The class conflict.** This failure is the known conflict between the RAR and the Solar-System quadrupole in this class of theories, already shown by Desmond, Hees & Famaey (2024). It is not new, and it is not a property of the turn-off. The statement is about the unscreened law only; the screened case is D7d.
- **Checks.** An exact limit derived for this lane, $q = -(16/49)\sqrt{e_N}$, is reproduced to $5 \times 10^{-10}$; two routes agree to $1.1 \times 10^{-3}$; our QUMOND value is 0.737 of Blanchet & Novak's AQUAL value (AQUAL fails harder). **K4 fails as frozen**: against Desmond et al.'s anchors our $|q|$ is 0.974, 0.965 and 0.965 of them (tolerance 3%); the gap is not a tail effect, and a lower $q$ only strengthens the failure. The script therefore exits 1 (7/8). The record's "tail" explanation of a 1.03 calibration factor is wrong.

## 3.3 Cassini $Q_2$ of the screened theory (D7d)

D7d solved the stiffened action (A33 (L) and (P), static, $O(\varepsilon)$) for the Sun in a uniform Galactic field, with a linear Fourier route and a nonlinear grid route. Its $L_\star \to 0$ limit is AQUAL with $\mu = x/(1+x)$: it reproduces Blanchet &

Novak's AQUAL value ($3.818 \times 10^{-26}$ against $3.8 \times 10^{-26}$). Criteria a13e418498, pre-result amendment e1ba64cf92, result b26f14bb82. CRISP FINDING; 10/10 checks.

- **The screened theory passes Cassini by $10^7$–$10^9$.** At $L_\star = 24\,\mathrm{pc}$, $Q_2 = 1.0$–$2.3 \times 10^{-34}\,\mathrm{s}^{-2}$ over the four fields and both footings; at 90 pc, 1.9–$4.3 \times 10^{-36}$. That is $2.2 \times 10^{-10}$ to $2.5 \times 10^{-8}$ of the Hees line. It passes Park et al. too.
- **Mechanism.** Inside $L_\star$ the Sun's **P** field is a constant hedgehog $GM/L_\star^2\,\hat{r}$, which has no quadrupole. $Q_2$ comes only from the anisotropic Galactic term acting at $r \sim L_\star$, so $Q_2 \propto GM/L_\star^3$. The stiffness is not a multiplicative screening factor in an external field.
- **The Cassini threshold.** The theory passes for $L_\star \geq L_{\star C}$ with $L_{\star C} \leq 0.069\,\mathrm{pc}$ (canonical) and $\leq 0.111\,\mathrm{pc}$ (alternative), at all four fields. This is an upper bound on the threshold, not a measurement: between 0.03 and 1 $r_M$ the solver did not converge, and $L_\star = 0$ fails. The binding lower gate stays CFG691's Saturn monopole (5.75 pc).
- **Consequences (reported only).** At $L_\star = 24\,\mathrm{pc}$ the inner-planet anomaly is $GM/L_\star^2 = 2.42 \times 10^{-16}\,\mathrm{m\,s}^{-2}$ (Mars at 0.17 of its bound), and a 20 kAU wide binary has $\gamma_v - 1 = 8.1 \times 10^{-6}$. The unscreened law-first reading predicts $\gamma_v \approx 1.17$–$1.20$ and fails Cassini. The two readings cannot both hold, and Gaia DR4 separates them.
- **Limits.** Uniform, static Galactic field over $\sim L_\star$; $O(\varepsilon)$; no ephemeris refit. The main run peaked at 3.0 GB, over the 2 GB budget (disclosed); the mutation run covers four checks (2/4, it bites). Record notes: CFG770's negative sign is the $\Phi$ convention; CFG832's 0.087 pc gate was an extrapolation of the right order.

### 3.4 The $Q_2$–RAR gate: does any unscreened law pass both? (D7c)

D7c scored the record's whole unscreened law family against both Cassini $Q_2$ (D7b's solver, Hees $2\sigma$, all four external fields) and SPARC (AG4b's per-galaxy pipeline, paired statistic, fail at $z \leq -2$). Criteria e9e5349368, result 26b5db080f. SCOPED NO-GO for the unscreened family, with one near-miss.

- **No law passes both on both footings.** The weakest tension anywhere in the family is $2.56\sigma$ (Hees), in one cell of the $\beta$ family.
- **Named laws.** All five named record laws (the canonical $n = 1$ law, $n = 2$, $\nu_{\mathrm{mono}}$, the standard $\mu$, $\nu_{\mathrm{RAR}}$) fail unscreened $Q_2$ at every external field on both footings: 1.18–4.15× the Hees line overall. All also fail Park et al.
- **The action's capacity family.** SPARC passes on both footings only at $n = 2$, whose $Q_2$ fails at 2.89–$8.24\sigma$. $Q_2$ passes at all four fields only at $n = 8$, which SPARC disfavours at $z = -6.32$ (canonical).
- **The near-miss.** In the $\beta$ family (the record's "effective description"), $\beta \geq 2.5$ passes $Q_2$ on the canonical footing and the graded SPARC statistic, but fails the alternative footing in the framework's own Milky-Way field by 1.19–1.23× for $\beta$ from 4 to 40, a floor no $\beta$ crosses. The secondary SPARC statistic disfavours the same members; which statistic one trusts decides the canonical reading.
- **External-field dependence.** The canonical law passes the Hees line only for a total external field $g_e \leq 8.20 \times 10^{-11}$ (canonical) or $8.13 \times 10^{-11}\,\mathrm{m\,s}^{-2}$ (alternative), below both the kinematic field at the Sun and the framework's own Milky-Way field.
- **Under AQUAL (E2) the no-go strengthens.** No member passes $Q_2$ at all four fields on both footings. The $Q_2$-only passer $n = 8$ now fails (alternative E3 cell, 1.04×). The $\beta$ members also fail the canonical E0 cell, by 1.01–1.12×: AQUAL raises their $Q_2$ by 1.03–1.22. The least-tense SPARC passer is $\beta = 5$, worst cell 1.41×.
- **Reconciliation with E1 (not papered over).** E1 redid D7c at three $a_0$ values, −1.4% to +3.0% around the canonical footing, with no alternative footing, and found that the $\beta \geq 3$ members pass Hees at all four fields and SPARC's primary statistic at all three $a_0$ (and $\beta = 2.5$ at two of three), at $Q_2$ of 0.87–0.99 of the line and SPARC $z$ of −1.59 to −2.00: D7c becomes UNDECIDED, with the secondary SPARC statistic still against them ($z \leq -5.39$). But E1 used QUMOND. E2's AQUAL raises the same members' $Q_2$ by 1.03–1.22, more than E1's margins, and at canonical $a_0$ they fail E0 by 1.01–1.12×. Neither lane computed AQUAL at E1's $a_0$ values. (E1 called these values native; they came from the background C2 shows is excluded, so v1.1 treats them only as an $a_0$ sensitivity bracket.) The combined status of the $\beta \geq 2.5$ members is therefore knife-edge on both sides: E1's $Q_2$ margins (0.87–0.99 of the line) are smaller than the AQUAL increase, and their SPARC passes sit within 0.41 of the $z = -2$ line. It is not a pass, and it is not a settled fail.
- **Checks.** 5/6: K4 fails as frozen (a tabulation difference of $3.3 \times 10^{-9}$ against a $10^{-10}$ tolerance; numerical only), and the script exits 1. The first run's K3 failure was check-side round-off, fixed post-result (amendment A1); that run is kept. The mutation run bites (3/6). The statements concern unscreened law forms only; the screened case is D7d.

# 4 The stiffness length $L_\star$ (D8, D8b, E4)

- **Two objects, one length.** A76's sound speed (at the cosmic mean density) is exactly AG6D's free-medium equation of state (0.720398 m/s at 24 pc), so AG6D's bound applies: in the free medium $\sigma_c \leq 1.70$ (canonical) or 1.87 km/s (alternative), below the $10^4$ K width by 4.84 and 4.41 times. $L_\star = \sigma_c^2/a_0$ as a free-medium stiffness is excluded; it survives only as a property of a tethered state that no action term yet defines.
- **Record error: the 24 pc floor is a measurement.** It is CFG820's globular-cluster $\chi^2$ bound (fine-grid roots 23.641 / 24.476 pc). It excludes the thermal $L_\star$: $\Delta\chi^2 = 3.913$ (canonical, by 0.07) and 5.992 (alternative). Erratum 9e6fbbbfcd.
- **Measured dispersions.** Six tidal dwarfs give median $L_\star$ 42.2 / 35.0 pc with an 84/16 spread of 4.45. Stilp et al. (2013), 23 dwarfs at a common 200 pc beam, give median $\langle\sigma_{m2}\rangle = 9.30$ km/s, so $L_\star = 29.95$ / 24.80 pc, with 84/16 = 1.995, a knife-edge pass of the frozen "one constant" line ($\leq 2$). The narrow-peak width gives 20.5 / 17.0 pc, below the floor.
- **But $\sigma$ is not universal (D8b).** Star formation does not drive it (FLOOR FAVOURED: $\rho = +0.474$, $p = 0.023$ against $\alpha = 0.0125$; slope +0.088, 95% CI [+0.026, +0.149], excluding 1/3). But its spread is real: $\chi^2 = 99.3$ for 22 degrees of freedom on tabulated errors, intrinsic scatter 0.96 km/s, against an 84/16 of 1.28 from noise alone. It tracks the H I column ($\rho = +0.613$, $p = 0.0024$). Read as a floor, the minimum $\sigma_{m2}$ gives 17.0 / 14.1 pc, below the globular bound on both footings. So D8b does not strengthen the conditional prediction $L_\star \approx 25$–30 pc: it holds only if $\sigma_c$ is a typical, gas-set value, and then $L_\star$ depends on environment, in conflict with the record's use of one $L_\star$ for gas-free systems.
- **The floor itself is soft (E4).** CFG820 built the 24 pc floor with the RAR fitting function and an additive external-field recipe, neither of which is the screened theory's physics. With the theory's own law and no recipe the floor is 28.85 pc (canonical) / 29.12 pc (alternative); with the law and the recipe kept, 23.87 / 24.66 pc, so the recipe, not the kernel, moves it. The action's own external-field response lies between and was not solved. The one stellar $M/L$ shifted by its $\pm 0.15$ dex spans 20.0–42.3 pc (canonical): the floor is calibration-dominated. Under the native model the thermal $L_\star$ is excluded at $\Delta\chi^2 = 6.21$ / 9.01. CFG821's dSph ceiling stays 22.0 pc, but it assumes no bound cold mass, which the screened theory requires for dwarfs; read literally the window [28.9, 22] pc would be closed.
- **Two verdicts above flip under clean inputs (E1).** D8's thermal exclusion ($\Delta\chi^2 = 3.91$, a 0.07 margin) does not survive an $a_0$ 1.4% below canonical (3.78, allowed; E1's "native" label for it is withdrawn in v1.1) or a $-0.2$ dex $M/L$ (0.32); D8b's floor reading flips both ways with the same $M/L$ shift. Neither is a firm verdict.

# 5 No covariant parent (D9, D9b, E5)

- **U(1)-dielectric parents (D9).** Any such parent that enforces the law's Gauss identity (phantom = cold medium) has a fastest speed $v_{\max}^2 \geq (1 + g/a_0)/\Lambda_{\rm LV}$, where $\Lambda_{\rm LV}$ is the dark sector's Lorentz-violation factor. It is exactly 1 for every Lorentz-invariant $L(X, Y)$, so all such parents are superluminal (the record's NG2, now for every $L(X, Y)$; erratum 05a3bfdb73). A foliated parent is causal only if $\Lambda_{\rm LV} \geq 1 + g/a_0$: $6.3355 \times 10^7$ (canonical) and $5.2478 \times 10^7$ (alternative) at 1 AU. The frozen exact-speed checks U2, U4 and U5 failed (a parity-odd term mixes polarisations; 9/12); a post-result amendment proved the bound survives the mixing (3/3). A map of all parent classes is in the lane (`MAP.md`).
- **The record's action escapes only by not enforcing the identity.** DSF-21's A33 action is causal because its field is the capped $u$; its cone excess at $\eta = \gamma$ is $1.19 \times 10^{-18}$ / $1.73 \times 10^{-18}$ at $L_\star = 24$ pc. Its identity is a matching, not a field equation.
- **Enforcing it (D9b).** With a multiplier enforcing $\mathbf{D} \cdot \mathbf{P} = \rho_c - \bar{\rho}$, the literal action counts the medium twice ($s = 2$): at $g = a_0/3$ it misses the law by $-0.55\, a_0$, with no deep branch. The law-preserving version is unique, has 4 second-class constraints and 7 degrees of freedom, is causal and well posed on FRW, and is exactly AQUAL with $\mu = x/(1 + x)$ in the static limit. But it meets every hypothesis of the no-self-gravity theorem D-T (growth $a^{0.60953}$ in EdS), and the positivity no-go NG1 applies ($\rho_c$ down to $-231\,\bar{\rho}$ at saddles, alternative). 14/14 checks; a first run with three code bugs (11/14) is kept.
- **With stiffness (E5).** The no-gos are a theorem for the stated classes, and they hold for the screened theory, not only at $L_\star \to 0$. At $L_\star \leq 90$ pc the stiffness moves the positivity no-go's saddle trough by at most $8.9 \times 10^{-4}$ (cancelling it would need $L_\star \approx 3.2$ kpc for the Milky-Way–M31 pair); the no-self-gravity theorem is independent of the stiffness; a dark-sector curl stiffness only raises the superluminal speed or adds a ghost; and no stiffness or parameter-free gate removes the collapse-gate ghost. The superluminality is a positivity violation of the law's own Lagrangian (an external argument the lane cites, arXiv:hep-th/0602178). Open classes: a foliated U(1)

under the owner's criterion that a global time function suffices (D9's row rests on $v \leq c$); non-local parents beyond CFG947; non-variational latches; dispersion-held media; multi-field latches; parents that do not enforce the identity. 14/15: the frozen check E1 fails (its pass line omitted cross terms of a truncated series); a post-result amendment shows the residual is exactly those terms.
- **Conclusion.** Enforcing the identity is what brings back D-T and NG1, with or without stiffness. No covariant parent is established; $L_\star$ and $\eta/\gamma$ remain inputs.

# 6 Tests and their power

## 6.1 The skew-sign test is retired (D1)

The record proposed the sign of the residual skew to separate $n = 1$ from $n = 2$. In $\log g$ both laws predict negative skew across the whole window ($\sqrt{N_s}$ skew at $a_0/2$: $-4.38$ for $n = 1$, $-2.44$ for $n = 2$). In linear $g$ the signs differ, but errors symmetric in $\log g$ make a linear skew: $+0.42$ from a 0.06 dex background, and 2.86 (canonical) / 3.18 (alternative) measured on SPARC outside the window, against best-case signals of $+0.062$ and $-0.164$ (canonical), 17–51 times smaller. The record's power estimate (7,000–13,500 points) also used the $N_s = 1$ skew with $N_s = 93.1$ normalisation; correcting it multiplies the required number by 93 ($n = 1$) and 384 ($n = 2$). SCOPED NO-GO on any rotation-curve data. Erratum f775a7e6c5.

## 6.2 The wide-binary fork is decided by systematics (D2)

- **1.18 is the law's number.** An independent FFT QUMOND solve gives 1.1971 for A24's case (A24: 1.1933) and, orientation-averaged at the pre-registered external field, 1.175 (canonical) / 1.210 (alternative). Medium-first predicts 1.000.
- **With the action's field equation (E2).** AQUAL lowers $\gamma_v$ by 0.007–0.011: 1.167 (canonical) / 1.199 (alternative) at the primary field. The registered Arm A band was built with QUMOND-type machinery, and the AQUAL values sit near its lower edge. A pre-data amendment A1 to the pre-DR4 note (ee621a0898) adopts the AQUAL values and tightens the scoring condition to a systematic below 0.048 (canonical) / 0.058 (alternative), not 0.054 / 0.064; it can only move outcomes to UNDECIDED. The frozen pre-registration is untouched.
- **The $\kappa$ leverage was overstated.** With the shared systematic and the external-field degeneracy, $\sigma(\kappa) = 0.073$, so the joint error goes $0.121 \to 0.077$, not 0.0605 (floor 0.048 at infinite $N$).
- **One bin cannot decide.** The DR3 frozen-cut catalogue has 43 of 6,210 pairs (0.69%) in 18–22 kAU, not A24's 7.41%. The whole-catalogue $\hat{\gamma}$ separates the law's floor from 1.000 at 5.04–5.85$\sigma$ (canonical) and 5.98–6.95$\sigma$ (alternative), but $3\sigma$ is unreachable at any $N$ if the systematic exceeds 0.054 (canonical) / 0.064 (alternative); the record's inter-group spread is 0.174. A pre-DR4 note fixes this rule before the data (commit 8658130bd0; the frozen pre-registration is untouched).
- **Satellites fail both readings.** A26's law-first verdict omitted the host's external field; with it, the 45 sub-threshold satellites sit +1.25 dex above the law (5 of 45 within 0.5 dex). Current data are open. Erratum c749c5007d.

## 6.3 The sealed ALMA tests are expected to be unscorable (D4)

The two governing plans (U4_27928, public 2026-11-06; GS4_24110, public 2026-11-28; dates from the record's archive metadata, not re-checked) are harness-complete on mock data. The record's forecast puts the gate statistic at S/N 0.18–0.58 (U4) and 0.04–0.15 (GS4) against a required 5, with no taper branch, so the expected outcome on both dates is "G1 FAIL, upper limit", with nothing scored. With perfect data, U4 reaches 4.34$\sigma$ (framework vs rival, governing kernel, canonical) only for $V_c \approx 240$–$260$ km/s and only while the 0.2 dex stellar-mass systematic is kept out; propagated, the separation is 0.00. GS4 cannot separate $n = 1$ from $n = 2$ ($\leq 0.51\sigma$ on every branch). The governing plans score only the $\nu_{\rm mono}$ kernel, so the canonical law has no governing reading; a kernel addendum is the owner's choice. One disc excludes no model.

### 6.4 The native $1024^3$ shear run died of memory (D3)

CFG960's pre-registered SC1x run (canonical footing) was killed four times by its own frozen swap rule, three times at the same force call (step 89 of 150, $a = 0.19$); the process footprint reached 65.2 GB on a 64 GiB machine, against a pre-registered estimate of 43–47 GB. SC2x (alternative) never started. No scored snapshot exists: **no verdict on either footing**. The chain stopped on 2026-10-22 at 01:01 local. Under the original frozen files a restart was expected to die at the same step. Since D3 (updated in v1.1): the ledger records the outcome (row CFG960-RUN-1021), and an owner-approved pre-result amendment A6 (f72bef68ea, committed alone) re-pins a lower-memory engine that is bit-identical to the frozen one on the lane's tests, with the step-90 replay at 13.4 GB above the particles (2ccd47be97); SC1x resumes from step 89. When v1.1 was written the chain had been restarted and no scored output existed: still no verdict on either footing.

## 7 The retained-fraction mechanism (D6)

The record's cluster lensing needs the law's phantom to supply a fraction $u = 0.4716$ of the observed excess inside splashback (P4). D6 looked for a mechanism. Criteria 3082c43c79, pre-score amendment daf92d2d11, result b21dda1a2d. CRISP FINDING, but no mechanism works.

- **The record's cluster model gives zero.** CFG959 sets its unrelaxed fraction to 1 for clusters, so P4's sourcing share is $1 - f_P = 0$, which is $13.0\sigma$ below 0.4716 (read the other way, $u = 1$, it is $14.6\sigma$ above). Getting 0.47 would need cluster galaxies with $v/\sigma = 1.64$. The two papers' "$u$" are complements under one symbol.
- **Record error: a sign.** At fixed retained fraction, $d\ln u/d\ln M_v = -1/6$ (numerically $-0.16680$), not the record's $+1/6$.
- **Stream counting gives $\frac{1}{2}$, and the profile rejects it.** If only inflowing cold matter sources the phantom (a new posit, not in the action), the self-similar infall solution gives $u \approx 0.50$ inside splashback at every accretion index tried, and $u_{\rm MID} = 0.489$ / 0.492 / 0.495 on the 318-cluster stack, within $0.7\sigma$ of 0.4716. But the 8-bin profile fails: $\chi^2 = 39.0$ / 37.3 / 36.7 for 8 degrees of freedom (canonical), the same as the record's own fitted 3D step, which the record already rejects. Its two second predictions also fail. Counting only the never-crossed stream gives $u_{\rm MID} = 0.06$–$0.14$.
- **Galaxy lensing gives a different $u$.** Isolated KiDS-1000 lenses give $u_{\rm gal} \geq 0.660 \pm 0.028$ (canonical) and $0.729 \pm 0.031$ (alternative), $6.8\sigma$ / $8.4\sigma$ above 0.4716 (statistical only; a $-0.25$ dex stellar-mass systematic would erase it). So $u \approx 0.47$ is not universal, but this exclusion does not survive clean inputs: with stellar masses $-0.2$ dex lower, $u_{\rm gal} = 0.501$, $+1.5\sigma$ (E1). Cluster hydrostatic masses give 0.344 or 0.230 depending on the bias assumption; the two ends differ by $6.1\sigma$, so this channel cannot yet test $u$ independently of lensing.
- **Disclosures.** K3 fails (the solver's interior slope is 0.625 against the asymptotic 0.75), so the deep profile is not certified and the script exits 1 (6/7). K4's tolerance was loosened after the run (an interpolation offset of $6 \times 10^{-6}$), disclosed in the code. The mutation run bites (5/7). Record error also noted: the record's "3D step in $r$" should read "a feature of the projected ratio".

## 8 Hidden inputs and the fitted constants (E1, E4)

Wave E applied the owner's data rule: every verdict compares raw measurements with the framework's own predictions, and ΛCDM-processed or standard-MOND ingredients are reported separately with the verdict's sensitivity to them.

- **Input ledger (E1).** 43 inputs, each classed raw, framework-native, survey calibration, ΛCDM-processed or standard-MOND machinery. Criteria 581d52210b, result 77a8b2d4a6; 6/6 checks.
- **The footings and the native $H_0$ (corrected in v1.1).** E1 took a framework-native $H_0$ of 55.0–57.5 km/s/Mpc from the record's $\beta = (2/3)\Omega_m$ background. That background is excluded by BBN (A78: $\beta < 0.027$, $G_{\rm cosmo}/G_N > 0.9611$; the $\beta$ branch has $G_{\rm cosmo}/G_N = 0.6845$ and $\Delta N_{\rm eff} = -3.43$). On the allowed background (A78's bound with CFG886's BAO + BBN fit) the native $H_0$ is 67.22–68.56 km/s/Mpc (lane C2, `covariant_push_2026_10_22/C2_aest_cosmology`, result 2dd388643a). Against the local ladder on disk ($73.0 \pm 1.0$, which SPARC's Hubble-flow distances use) that is $4.44$–$5.78\sigma$ (ladder error only) or $3.93$–$5.15\sigma$ (native error added): the ordinary Hubble tension, not the 15–$18\sigma$ of v1.0. The canonical $a_0$ is unaffected: $\kappa c\sqrt{G\rho_\Lambda}$ is the same number on either background. Withdrawn, because they rested on the excluded background: E1's check that the canonical footing is "clean to 3%" (it read $a_0 = cH_0/Z$ with that native $H_0$), and E1's statements

that the alternative footing has no native counterpart and is a pure ΛCDM import. C2 gives no corrected number for these; which footing a native $H_0$ supports is open again.

- **Verdicts that flip.** Four do not survive clean substitution: D7c (UNDECIDED in E1's $a_0$ bracket, with the reconciliation in Section 3; the bracket's "native" label is withdrawn in v1.1), D8's thermal-$L_\star$ exclusion, D8b's floor reading, and D6's "$u_{\rm gal}$ excludes 0.47". The others survive (D1, D2, D4, D5, D6 items 1–3, D7, D7b, D8b items 1–2).
- $\kappa$ **(E4).** With a ΛCDM-free background (BAO + BBN) the gas routes give $a_0 = 7.986 \times 10^{-11}\,\mathrm{m\,s^{-2}}$ and $\kappa = 0.414 \pm 0.094$ (canonical), a shift of $-0.03\sigma$: $\kappa$ is robust to the background. What the data fix is $a_0$; writing it as $\kappa$ adds the claim $a_0 \propto \sqrt{\rho_\Lambda}$. Criteria 9cc8fcae21, pre-result amendment 9b3dfb1000, result c5a9768e6a; 6/6.
- $u$ **(E1, E4).** $u$ depends on the distance convention and on $a_0$: $d\ln u/d\ln a_0 = +0.55$. Through the distance convention it moves with $h$: 0.497 at $h = 0.674$ and 0.444 at $h = 0.730$ (E1). (v1.0 quoted a value at E1's native $h$; that $h$ came from the excluded background, C2 gives no replacement, and the number is withdrawn.) At the galaxy-measured $a_0$, $u_{\rm MID} = 0.433 \pm 0.030$ (post-result, reported only), and the error $a_0$ carries into $u$ (0.059) is the largest term in its budget. So $u$ is not an independent measured fraction.
- **Independence.** The three constants are independent in the action (Jacobian rank 3, singular values 1.05, 0.62, 0.35); no candidate relation among them is forced, so there is no second prediction to test.

## 9 Dwarf satellites against the screened theory (E3)

Criteria 5686635887, result c5e4196bb9; 6/6.

- **The excess is real.** The 45 sub-threshold satellites sit $+2.01 \pm 0.12$ dex above Newton with statistical binary corrections ($+1.64 \pm 0.12$ in the worst clean combination).
- **But +2.15 dex is the wrong yardstick.** At $L_\star = 24$ pc the screening still gives these systems a median boost of $10^{1.23}$ over Newton. The data sit $+0.89 \pm 0.13$ dex above the screened prediction (alternative +0.85), and +1.52 dex at $L_\star = 90$ pc. The record's "largely screened and nearly Newtonian" is wrong.
- **The capacity is undecidable.** At the census retained fraction, the median system must hold $q = 0.81$ of its whole catchment's cold energy inside its half-light radius; census-free, $q = 8.0$, a shortfall of +0.90 dex. The census is ΛCDM-processed and the record holds no binding mechanism. UNDECIDABLE on both footings.

## 10 Record corrections

Six errata were written fix-forward into the record (originals kept): f775a7e6c5 (D1: skew-sign discriminator retired), e1fd7be4dc (D5: the data do not select $R = 4$), 3c11f5d2e0 (D7: turn-off needed; planet margin; thermal $y_t$), 9e6fbbbfcd (D8: the 24 pc floor is CFG820's bound; free-medium $L_\star$ excluded), c749c5007d (D2: $\kappa$ leverage, bin occupancy, fork not resolved), 05a3bfdb73 (D9: NG2 covers all Lorentz-invariant $L(X, Y)$). The DSF-21 documents now call $\kappa = Z/4\pi$ a postulate and credit Milgrom (d3e4110528).

## 11 What this establishes, and what it does not

- **The screened theory.** It is the one version that passes both Cassini (by $10^7$–$10^9$) and the galactic law. Its open items: the dwarf gap (+0.89 dex, capacity undecidable), a calibration-dominated $L_\star$ floor, an $a_0$- and $h$-dependent $u$, and no covariant parent in the stated classes. Its decisive test is Gaia DR4: $\gamma_v - 1 \approx 8.1 \times 10^{-6}$ against the unscreened reading's $\approx 1.17$.
- **Established.** $\kappa$ is not derivable in the horizon-thermodynamic class, and the data do not select $\frac{1}{2}$ among natural caps. The unscreened canonical law fails Cassini $Q_2$ by 2.44–5.19× with the action's AQUAL field equation (1.7–4× in QUMOND), and the turn-off cannot change that; no member of the unscreened law family passes both $Q_2$ and SPARC on both footings. The turn-off has no mechanism in the action. Free-medium $L_\star$ is excluded. Gauss-enforcing parents are superluminal, Lorentz-violating, or lose self-gravity. The skew-sign test is retired. No tested mechanism gives the retained fraction $u \approx 0.47$, and it is not universal.
- **Open and data-gated.** The wide-binary fork (Gaia DR4, if the systematic is below 0.048 / 0.058 after the AQUAL amendment). The ALMA tests (expected unscorable). The native shear test (no verdict). The $\beta$ members of the unscreened family (knife-edge between E1 and E2). The dwarf capacity (needs a native

catchment rule or isolated dwarfs).

- **Not derived.** $\kappa = \frac{1}{2}$, $L_\star$, $\eta/\gamma$, the retained fraction. No covariant parent is established. Cold energy remains a medium, not a particle species. No result here is evidence for the framework over its rivals.

**Disclosures.** D7b's frozen check K4 fails (anchor convention; exit 1) and one post-result amendment explains it. D9's frozen checks U2, U4, U5 fail; amendment A1 (post-result, 7bdd5fee57) adds the lower-bound checks; the first run is kept. D9b's first run (11/14, code bugs) is kept. D8's amendment A1 (ce9340ccbc) and A2 (95a7ec3a41, the owner-approved fetch of Stilp et al.) are pre-result. D8b's K4 label parser was fixed after its first run (no criterion changed). D7c's K4 fails as frozen; its amendment A1 is post-result (check-side round-off) and its first run is kept. D7d's amendment A1 is pre-result; its main run used 3.0 GB (over budget) and its mutation run covers four checks. E1's first run misparsed its ledger header (fixed before the scored run). E3's mutation run passes K6, against the criteria's expectation (disclosed by the lane). E4's $u$ at the galaxy $a_0$ is post-result. E5's frozen check E1 fails; its amendment A1 is post-result. D6's amendment A1 is post-prototype and pre-score; its K3 fails and its K4 tolerance was loosened after the run. D3 is read-only forensics with no criteria file. D1's best case is an upper bound on the signal, not a forecast. D4's power numbers are noise-free ceilings. Not blind: the lanes read the record's numbers before freezing.

**Changes in v1.1 (2026-10-22).** One correction, approved by the owner, from lane C2 (`covariant_push_2026_10_22/C2_aest_cosmology`, result 2dd388643a) and the record's A78 BBN bound (dc7cc7e409). (1) The framework-native $H_0$: 67.2–68.6 km/s/Mpc, 4.4–5.8$\sigma$ (ladder error) or 3.9–5.2$\sigma$ (combined) below $73.0 \pm 1.0$, replacing v1.0's 55.0–57.5 and 15.5–18$\sigma$, which came from the $\beta = (2/3)\Omega_m$ background that A78 excludes. (2) Withdrawn as resting on that background: the "canonical footing clean to 3%" check, the "alternative footing is a pure ΛCDM import" statement (abstract and Section on hidden inputs), the value of $u$ at the native $h$, and the "native" label of E1's $a_0$ bracket in the D7c reconciliation and the D8 flip. (3) Status update: CFG960's amendment A6 and restart (Section on D3). (4) The canonical $a_0$, every Cassini, wide-binary, dwarf, parent and $L_\star$ result, and the D7c and D8 flips as $a_0$- or $M/L$-sensitivity statements are unchanged.

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Repository lanes (criteria, scripts, outputs) under `doors_push_2026_10_22/`: D1–D9b as in Table 1; the pre-DR4 note under `prep_2026/gaia_dr4_prep/`. Companion notes: PAPER49 v1.2 (doi:10.5281/zenodo.23533323), PAPER55 (doi:10.5281/zenodo.23482936), PAPER56 (doi:10.5281/zenodo.23532875), PAPER57 (doi:10.5281/zenodo.23532885). Planetary bounds: A. Hees, W. M. Folkner, R. A. Jacobson, R. S. Park, PRD 89, 102002 (2014); R. S. Park et al., PRD 114, 024066 (2026). Class conflict: H. Desmond, A. Hees, B. Famaey, MNRAS 530, 1781 (2024). Cross-check: L. Blanchet, J. Novak, arXiv:1105.5815 (2011). Data: S. A. Cookson et al., MNRAS 547, stag342 (2026) (wide binaries); A. M. Stilp et al., ApJ 765, 136 (2013) (dwarf H I dispersions); F. Lelli et al., A&A 584, A113 (2015) (tidal dwarfs); F. Lelli, S. McGaugh, J. Schombert, AJ 152, 157 (2016) (SPARC). Theory: M. Milgrom, ApJ 270, 365 (1983); M. Milgrom, Phys. Lett. A 253, 273 (1999) (astro-ph/9805346); M. Milgrom, arXiv:1605.07458; M. Milgrom, MNRAS 403, 886 (2010) (QUMOND). AI assistance: Anthropic's Claude (lanes, review, drafting).